THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. While adding two numbers digit by digit, the carry from any column can never exceed 1 because: 1

(a) the digits are always less than 5
(b) the largest possible column total is 9 + 9 + 1 = 19
(c) carry is always reset to 0
(d) the numbers are written in base 5

2. At the start of the algorithm for the divisors of n, the list-of-divisors is: 1

(a) empty **(b)** [n] **(c)** [1] **(d)** [1, n]

3. The ‘long division’ method of finding the gcd using Indian place-value numerals is found in the work of: 1

(a) Al-Khwārizmī **(b)** Euclid **(c)** Āryabhaṭa **(d)** Archimedes

4. **Assertion (A):** The simplest gcd algorithm that scans every j from 1 to min(m, n) takes 10 times as much work when the smaller number goes from 3 digits to 4 digits. 1
Reason (R): The work of that algorithm is proportional to the digits of the numbers.

(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true but R is not the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

5. While adding 2764 and 3589 by the column algorithm, the number of columns that produce a carry of 1 is: 1

(a) 3 **(b)** 5 **(c)** 4 **(d)** 2

6. A quadrilateral is convex when: 1

(a) one internal angle is more than 180°
(b) its diagonals do not intersect
(c) its sides are all equal
(d) all its internal angles are less than 180°

7. If the diagonals of a quadrilateral bisect each other, then it is a: 1

(a) trapezium **(b)** kite **(c)** rectangle **(d)** parallelogram

8. A line through the midpoint of one side of a triangle and parallel to another side: 1

(a) is equal to the third side **(b)** bisects the third side
(c) is perpendicular to the third side **(d)** meets the third side at a vertex

9. The Varignon parallelogram of a quadrilateral whose diagonals are equal and perpendicular is: 1

(a) a trapezium **(b)** a rectangle that is not a square

(c) a square **(d)** a rhombus that is not a square

10. **Assertion (A):** A quadrilateral with one pair of equal opposite sides is always a parallelogram. 1
Reason (R): In a parallelogram both pairs of opposite sides are equal.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State Euclid's observation used to reduce gcd(m, n), where m ≥ n. 2

12. Apply Euclid's subtraction algorithm to gcd(18, 12). Write the sequence of pairs. 2

13. True or false, with a reason: a parallelogram with one right angle is a rectangle. 2

14. True or false, with a reason: a rhombus whose diagonals are perpendicular is a square. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Use the division algorithm (long division) to find gcd(825, 375) and state the number of reductions. 3

16. The sides of $\triangle ABC$ are AB $= 14$ cm, BC $= 18$ cm and CA $= 10$ cm. D, E, F are the midpoints of AB, BC, CA. Find the perimeter of $\triangle DEF$ and the length of each of its sides. 3

17. Find gcd(2430, 1215) by the division algorithm. 3

OR

Verify, for m = 33 and n = 7, that the common divisors of m and n are the same as the common divisors of n and m mod n. 3

18. ABCD is a parallelogram. P and Q lie on diagonal BD so that $DP = BQ$. Show that APCQ is a parallelogram. 3

OR

In $\triangle ABC$ the medians are AD, BE and CF meeting at G. If AD $= 24$ cm and BE $= 30$ cm, find AG, GD, BG and GE. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Write an algorithm prime(n). Execute it for n = 21 and n = 23. 5

OR

Write an algorithm primedivisors(n). Execute it for n = 60. 5

20. In quadrilateral ABCD, $AB \parallel DC$ and $AB \neq DC$. G and H are the midpoints of AC and BD. Prove that $GH \parallel AB$ and find GH when AB $= 12$ cm and DC $= 20$ cm. 5

OR

Show how to tile the plane with copies of any given triangle ABC, using parallelograms. 5

END OF QUESTION PAPER — **Total: 40 Marks**